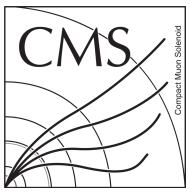
Internal review draft – 20 October 2026
No CMS DP number assigned



The Compact Muon Solenoid Experiment
CMS Performance Note

MCP-referenced timing calibration of CMS BTL prototype bars

CMS

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Abstract

A timing-calibration procedure is studied using CMS barrel timing layer (BTL) prototype scintillator bars in the September 2025 beam test. A microchannel-plate photomultiplier (MCP-PMT), digitized by an oscilloscope, provides an external arrival-time reference. A trigger link identifies coincident events in the BTL readout and the oscilloscope. Segment-dependent clock alignment and a locally weighted smoothing correction are used to map the reference into the BTL time coordinate.

The dependence of the measured arrival time on pulse charge is described by quadratic functions of inverse charge for four silicon-photomultiplier channels on two reference bars. These functions provide an empirical time-walk calibration over the measured charge range. Timing-difference distributions illustrate the validation strategy and the need to distinguish intrinsic reference-detector resolution from residual synchronization effects. This draft documents the available results and identifies the remaining selection, uncertainty and reproducibility checks before publication.

Draft v1 for scientific discussion. Not approved for public release.
The following slides retain explicitly marked review items.

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A timing-calibration procedure is studied using CMS barrel timing layer (BTL) prototype scintillator bars in the September 2025 beam test. A microchannel-plate photomultiplier (MCP-PMT), digitized by an oscilloscope, provides an external arrival-time reference. A trigger link identifies coincident events in the BTL readout and the oscilloscope. Segment-dependent clock alignment and a locally weighted smoothing correction are used to map the reference into the BTL time coordinate.

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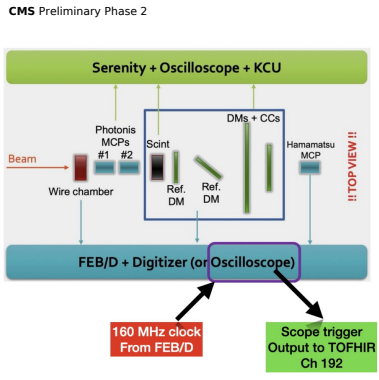
Beam-test setup and the reference signal



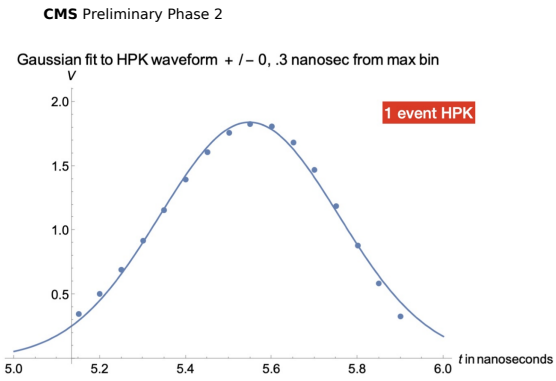
The September 2025 beam test used scintillator bars read out at both ends by silicon photomultipliers (SiPMs). A downstream MCP-PMT provides a fast reference for a restricted region of the bars.

The BTL front-end readout and oscilloscope use different time coordinates. The scope trigger reaches TOFHIR channel 192; a separate 160 MHz clock link can also be recorded by the scope.

The highlighted lower readout path is used for the reference-bar calibration.

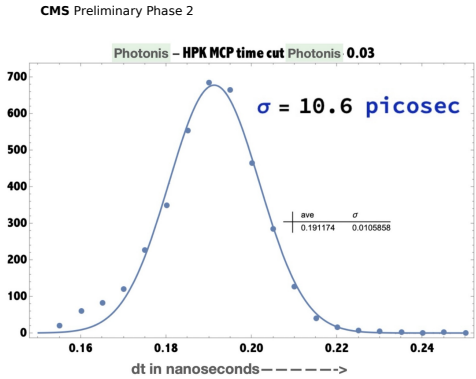
Legacy setup illustration from the supplied Note1 source. Acronyms inside the illustration will be simplified in the final redraw.

Reconstructing the MCP pulse time



An example Hamamatsu MCP waveform is shown with its fitted pulse template. The fit estimates the pulse centroid rather than using the threshold-crossing time alone, reducing sensitivity to pulse amplitude.
The scope-trigger time identifies the event; the fitted peak supplies its relative reference time. Their difference is measured within the same oscilloscope waveform.

Intrinsic reference timing is a separate measurement

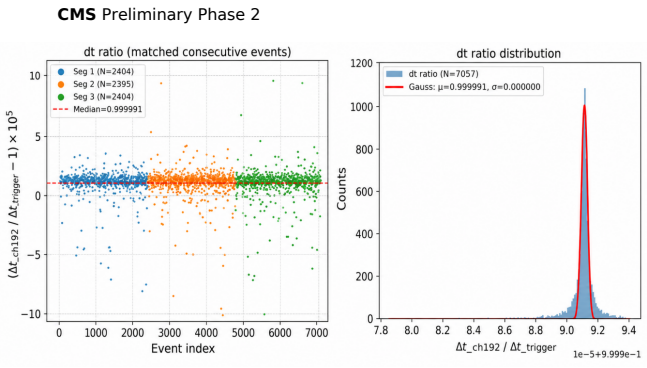


The distribution compares the fitted times of two MCP-PMTs recorded by the oscilloscope. The measured width is 10.6 ps.

Assuming independent errors and a 7.5 ps PHOTONIS contribution gives $\sigma_{\mathrm{HPK}} \simeq \sqrt{10.6^2 - 7.5^2} = 7.5$ ps.

This scope-only estimate does not include synchronization to the BTL readout.

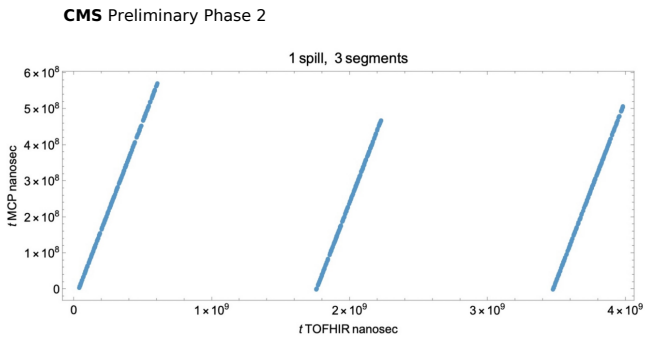
Matching the same particles between readouts



Consecutive trigger intervals measured by channel 192 are compared with those recorded by the scope. Correlated intervals identify candidate matches despite the independent clocks.
The interval-ratio distribution tests matching consistency. It is not a measurement of the picosecond resolution of the reconstructed particle time.
Only uniquely matched events enter the reference calibration.

Implementation (repository-relative):
`FEBD_Sync/simple_sync.py`
`dedup_triggers`: duplicate removal.
`match_cumulative_time`: matching and tolerance.

Clock-rate alignment within acquisition segments

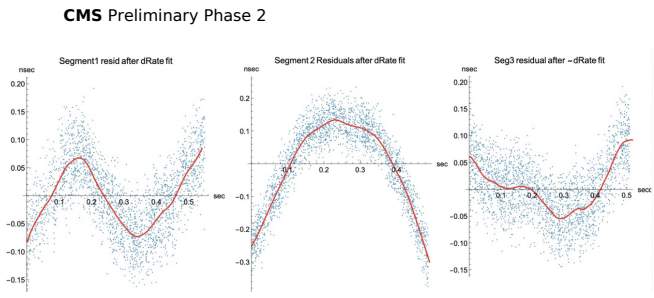


Matched scope-trigger times are compared with TOFHIR trigger times. A nearly linear relation describes the overall clock-rate difference within an acquisition segment.

An offset and a slope are needed: a slope close to one is not sufficient over a long time interval. Clock resets require separate segment coordinates.

Legacy clock-relation plot. This plot illustrates synchronization, not charge-dependent time walk.

Correcting smooth residual clock drift



After the linear clock-rate correction, the time residual still changes within each of three segments. Points are matched events; red curves show the smooth trend estimated by LOWESS, a locally weighted regression.
The fitted trend is added to the linear prediction to reconstruct the expected channel-192 trigger time. This expected time is subtracted once from the SiPM time in the reference difference.

Legacy residuals in ns versus segment time in seconds. These are pre-LOWESS residuals, not final timing resolutions. Implementation: `TimeCalibration/ch192_vs_trigger_lowess.py`, lines 657–668 and 816.

A single, explicit time convention

Let $g(u)$ map a scope-trigger timestamp u into the TOFHIR time coordinate, including the linear clock relation and the fitted smooth drift. Define the within-waveform difference

$$\delta_{\mathrm{MCP}} = t^{\mathrm{scope}}_{\mathrm{peak}} - t^{\mathrm{scope}}_{\mathrm{trigger}}.$$

The reference difference implemented in the calibration is

$$\Delta t_i = t_i^{\mathrm{TOFHIR}} - g(t^{\mathrm{scope}}_{\mathrm{trigger}}) - \delta_{\mathrm{MCP}},$$

up to a fixed cable/geometry offset and the negligible-rate-conversion approximation for the short within-waveform interval.

The charge-dependent component of Δt_i is the time walk. An arbitrary constant does not change a shape calibration.

Implementation: `TimeCalibration/ch192_vs_trigger_lowess.py`, lines 816 and 1102. Both per-file and combined-run calculations subtract (`mcp_peak - mcp_trig`).

Reference bars and calibration selection

Upstream bar	SiPM channels	Role
7	133, 154	First reference pair
8	137, 150	Second reference pair

- Select coincident SiPM hits and a unique MCP-associated event, using channel 192 as the TOFHIR indicator.
- Require a successfully reconstructed MCP pulse and consistent peak-to-trigger timing. This is a waveform-quality cut, not a particle transit-time measurement inside the MCP.
- Use Q, the TOFHIR ADC charge proxy. It is not a deposited energy in MeV.
- Fit the profile of Δt_i versus $1/Q$ over the retained charge range.

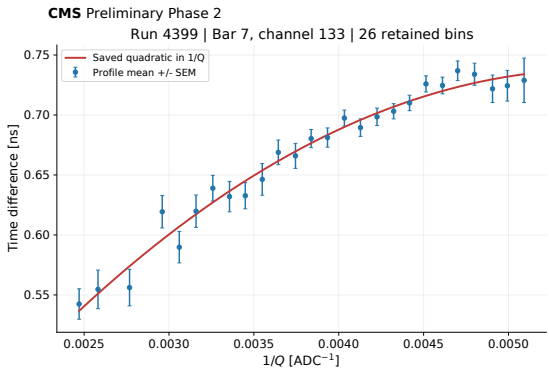
Implementation (repository-relative):

`TimeCalibration/ch192_vs_trigger_lowess.py`: coincidence and walk-sample selection.

`TimeCalibration/ch192_vs_trigger.py`: `build_mcp_map`, optional MCP cuts.

`TimeCalibration/bar_processing.py`: peak-to-trigger timing selector.

Bar 7: channel 133 reference calibration

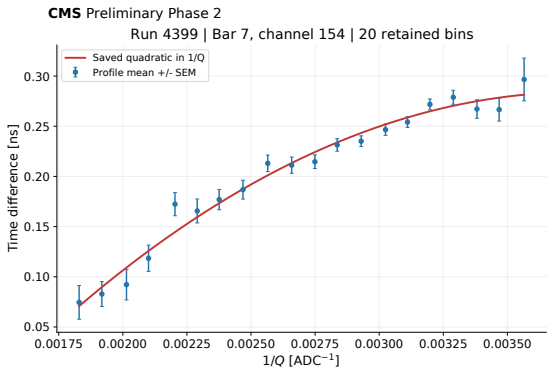


Run 4399 events are grouped in inverse-charge bins. Each point is the mean time difference; the vertical bar is its saved standard error of the mean (SEM). The curve is the saved binned quadratic $W(Q) = a + b/Q + c/Q^2$. There are 26 retained profile points. The time coordinate includes the historical constant display offset.

Review N1-4: Confirm the historical fit weighting and clipping. The plotted SEM does not by itself imply weighted fitting.

Redrawn from archived profile and coefficient CSVs. No refit. Saved display convention includes a +26 ns shift.

Bar 7: channel 154 reference calibration



The opposite end of bar 7 is calibrated independently. Points show the mean time difference and SEM in 20 retained inverse-charge bins.

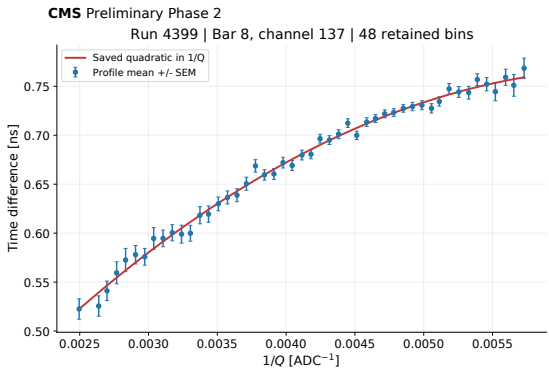
The red curve is the saved quadratic in $1/Q$, not a quadratic in Q.

Channel-dependent coefficients allow for different pulse responses at the two bar ends.

Review N1-4: The manuscript's common 12-bin statement is incompatible with these saved profile tables.

Run 4399 legacy calibration, redrawn without refitting. Same +26 ns display convention as channel 133.

Bar 8: channel 137 reference calibration



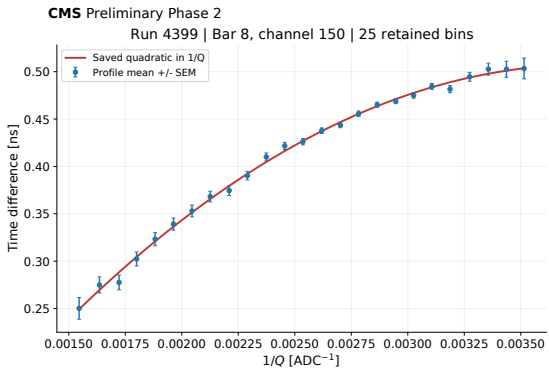
The channel-137 time difference increases as inverse charge increases: smaller signals cross a fixed threshold later.

Points are profile means with SEM; the red line is the saved binned quadratic. There are 48 retained points. The correction is applicable over the measured charge support, not automatically outside it.

Review N1-4: Retain this legacy result or replace all four channels with one consistently regenerated calibration.

Run 4399 legacy calibration. Coefficients and numerical points are included with the draft.

Bar 8: channel 150 reference calibration



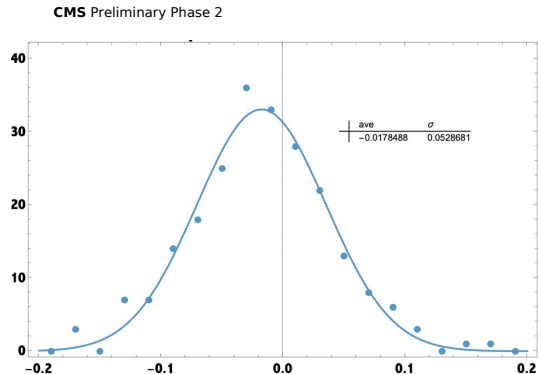
The opposite end of bar 8 has its own charge-dependent delay. The 25 points show the mean timing difference and SEM; the curve is the saved binned quadratic.

The four curves demonstrate the calibration procedure. Different numbers of retained points are preserved rather than imposing an undocumented common binning.

Review N1-4: The newer reference used for the extended comparison in Note2 has different selections and fit weighting.

Run 4399 legacy calibration. Coefficient units are ns, ns ADC and ns ADC^2 for a, b and c.

Timing validation using the two bar ends



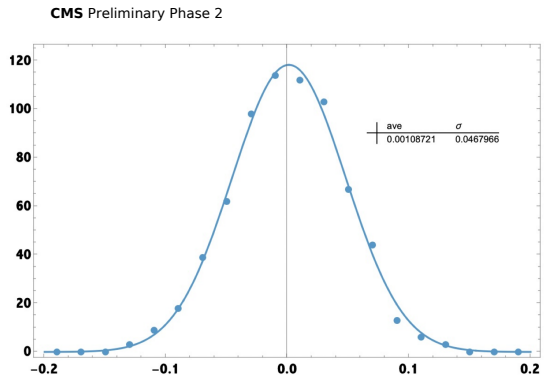
The distribution is the corrected time difference between the two SiPM ends of bar 8. The displayed Gaussian width is approximately 53 ps.

For uncorrelated end errors and controlled hit-position effects, $\sigma((t_L + t_R)/2) = \sigma(t_L - t_R)/2$, giving roughly 26 ps. Common timing jitter cancels in the end difference.

Review N1-5: Recover histogram inputs, fit range, uncertainty and position selection before treating 26 ps as a final resolution.

Legacy timing distribution. The inference is conditional; it does not measure synchronization jitter.

Timing validation against the MCP reference



The corrected bar-average time is compared with the MCP time. The legacy panel reports a width of 46.8 ps. This difference includes the bar, MCP and residual synchronization contributions, including any correlations. It must not be identified with the intrinsic MCP resolution.

Review N1-5: Establish whether 46.8 ps is a Gaussian-fit width or sample RMS, and recover its uncertainty and event selection.

Legacy bar–MCP distribution, not refitted for this draft. Do not combine unrelated selections or clock branches in quadrature.

Reference-calibration result and scope

- A trigger link permits common-event calibration across independent BTL and oscilloscope time coordinates.
- A linear clock relation plus a smooth residual correction defines the reference-time mapping.
- Four saved inverse-charge profiles and quadratic functions describe the reference-bar time walk. Numerical points, SEM and coefficients accompany this draft.
- Intrinsic MCP timing, reference-bar timing and cross-system synchronization are different performance quantities.
- Charge range and operating conditions limit transfer to other samples; a constant time offset is not a walk-shape difference.

The present evidence supports the calibration workflow. Final precision claims require a frozen selection and complete statistical/systematic treatment.

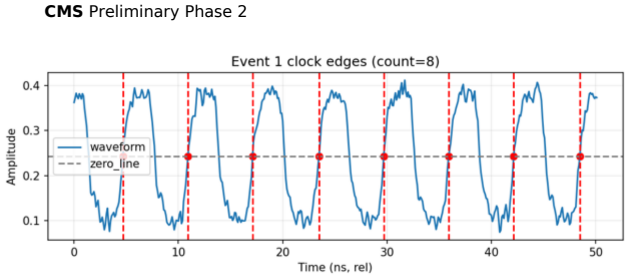
Review queue before publication

ID	Decision or missing evidence
N1-4	Choose legacy or updated calibration; align bins, clipping, weighting and coefficient convention.
N1-5	Reproduce bar timing widths with uncertainties and position/covariance assumptions.
N1-6	Verify citations, official template/DP metadata and final publication packaging.

Sources: supplied `Note1.pdf`; candidate editable manuscript `CMS-Note/cmsnote`; archived calibration CSVs in `Q_Walk/reference_quadratic_t0`.

The figure manifest records the exact copied inputs and checksums. This review slide is removed from a public release.

Backup: clock-phase reconstruction



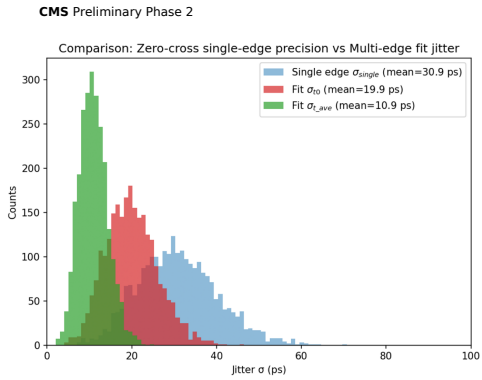
The reference clock is sampled by the oscilloscope. Several rising-edge crossings can be reconstructed within a waveform.

Fitting edge time against edge index allows the clock phase to be estimated using multiple cycles. Evaluating the fit near the mean edge index avoids unnecessary extrapolation.

This is an alternative synchronization diagnostic, distinct from the trigger-link branch used for the legacy reference curves.

Legacy illustration. Correlated edge errors prevent assuming an automatic $1/\sqrt{N}$ precision improvement.

Backup: comparing clock estimators



The recorded MCP time is compared with three clock-phase estimators: a single edge, a regression intercept and the fitted time at the mean edge index. The distributions illustrate estimator dependence. Their widths contain the contributions present in this specific measurement and are not automatically an absolute clock-accuracy measurement.

Review N1-6: Recover source arrays and confirm common selections and width definitions before using numerical improvements.

Legacy clock-estimator diagnostic, retained as a discussion backup.

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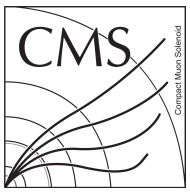

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A charge-dependent time-walk model is studied for CMS barrel timing layer (BTL) prototype channels using two discriminator thresholds. Their current separation divided by the measured crossing-time separation provides a finite-interval pulse-slope proxy. A linear profile fit describes this proxy as a function of the measured charge, and an inverse-slope function models the timing correction with one amplitude parameter per channel.

Slope measurements from run 4076 are compared with independent microchannel-plate (MCP) reference calibrations from run 4399. The two runs contain different events. A free constant offset and the correction amplitude are fitted only where both measurements have populated charge support. The resulting agreement is a calibration-shape comparison, not an independent measurement of timing resolution.

The draft also examines channel-to-channel variations and an exploratory prediction of the correction amplitude from internal pulse information. Such a prediction could enable operation without an event-by-event external time reference, but its transferability and full uncertainty remain to be established.

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Two complementary runs, not the same events

Sample	Available information	Role in this study
Run 4076	Valid dual-threshold timing and charge	Measure the pulse-slope proxy $S(Q)$
Run 4399	MCP-associated reference timing and charge	Measure the empirical walk $W_{\mathrm{MCP}}(Q)$

The runs contain different particles and are never matched event by event. The comparison transfers a channel-response model between runs.

The upstream and downstream modules each contain bars read out at two ends. The channel/bar mapping determines the coincidence pairs; the same channel number is compared across the two runs.

Review N2-1: Document gain, bias, thresholds, temperature, firmware and charge calibration to justify cross-run transfer. The newer run-4399 reference is not the legacy four-curve version displayed in Note1.

From two crossing times to a walk model

For thresholds $I_1 < I_2$ and their valid crossing times t_1, t_2, define

$$S = \frac{I_2 - I_1}{t_2 - t_1} \quad [\mu\mathrm{A/ns}], \qquad \overline{S}(Q) \simeq c_0 + c_1 Q.$$

S is a finite-interval rising-edge slope proxy. It is not the current difference ΔI itself and is not necessarily the instantaneous derivative at the first threshold.

$$W_Q(Q) = \frac{A}{c_0 + c_1 Q}, \qquad t_{\mathrm{corr}} = t_{\mathrm{measured}} - W_Q(Q).$$

The amplitude A (AWC) has units of current and is constant for a channel and operating configuration. The correction varies event by event through Q.

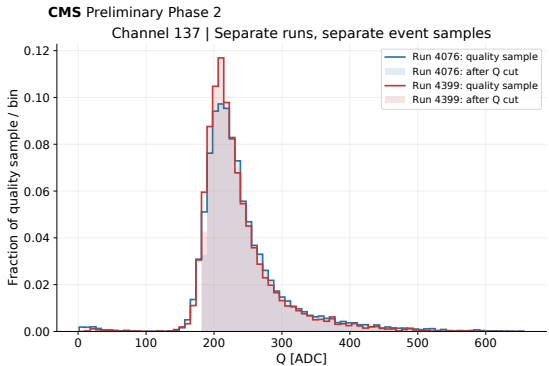
Review N2-2: Freeze the safe-t_2 decoding, threshold conversion and physical timing cuts. This is an effective pulse model, not an exact waveform identity.

Selection and profile construction

- Run 4076: require exactly the two hits of the target bar in its module. Hits in the other module are not constrained; channel 192 is ignored.
- Retain valid rising-edge timing and the selected prompt-slope population, removing unphysical and slow-slope events before profiling.
- Retain charge above the lower half-maximum boundary of the fitted Landau–Gaussian peak. The selected data extend to the right of the peak.
- Form ten charge bins. Plot the mean S and the hit standard deviation (SD); use the standard error of the mean (SEM) for the linear-fit weights.
- Bootstrap the selected hits with fixed charge-bin edges and refit. The central 68% interval uses the 16th and 84th percentiles of 1000 fits.

Review N2-2: Use the frozen extended-profile metadata. Do not silently inherit newer code defaults. These full-statistics profiles are not train/test splits.

Charge spectra define the available comparison sample

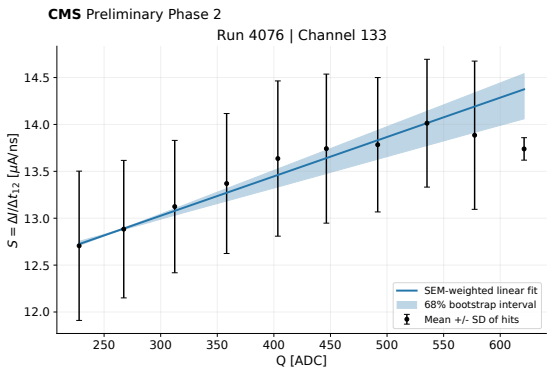


Channel 137 charge spectra are shown for the two independent runs. Lines show quality-selected hits; shaded spectra retain the additional charge-peak selection. Each run is normalized to its own quality-sample count.

The charge proxy is in ADC units, not MeV. Different populations and acceptances motivate restricting the curve alignment to jointly supported charge values.

Recovered histogram counts, not a new peak fit. Run-4076 profile selection starts at about 184.9 ADC for this channel.

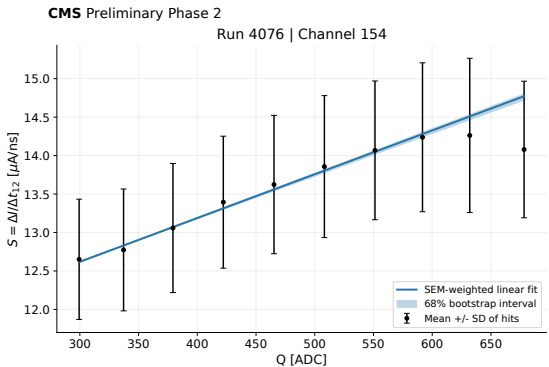
Channel 133: fitting the slope profile



Run 4076 channel-133 hits are grouped into ten charge bins. Points show mean S; bars show the SD of individual selected hits. The line is the saved SEM-weighted fit $c_0 + c_1 Q$.
The shaded interval is the central 68% bootstrap interval of the fitted mean curve. It is shown at its true size, without visual enlargement.

Extended-profile snapshot. SD describes hit spread; SEM and the fit band describe different statistical quantities.

Channel 154: the opposite end of bar 7

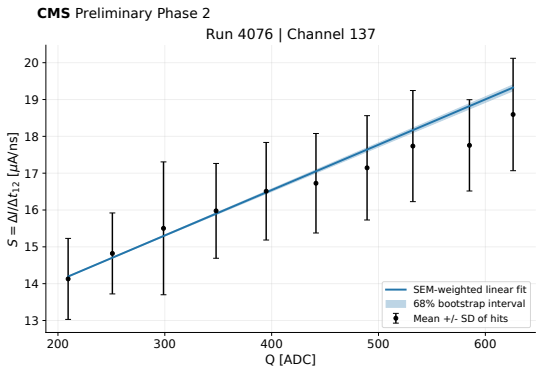


The same selection and profile procedure is applied to channel 154. The mean slope increases with charge, but the response is calibrated independently at each bar end. Bars show hit SD. The line uses SEM weights and the band propagates the saved bootstrap coefficient replicas, including their correlation.

Review N2-3: Assess linear-model adequacy separately from the statistical band; bootstrap spread is not a model-bias uncertainty.

Run 4076, extended-profile snapshot. No model refit is performed for the slide rendering.

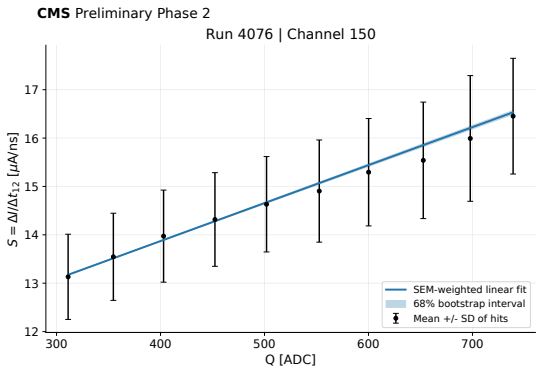
Channel 137: pulse slope and charge



The selected run-4076 sample contains 19 126 channel-137 hits. Ten profile points constrain a linear pulse-slope model over the observed charge range. The SD bars show event-to-event spread, while SEM weights determine the fit. The narrow shaded band is the conditional 68% bootstrap interval of the mean prediction, not a single-hit uncertainty.

Selected count and fit are read from the saved extended-profile metadata. No uncertainty band is magnified.

Channel 150: the opposite end of bar 8



Channel 150 is treated with the same profiling procedure. The charge-dependent slope provides the denominator of the proposed walk function; a separate amplitude A determines the correction strength. The line and shaded interval describe the fitted mean. Points and SD bars retain information on the spread within the selected charge bins.

Run 4076, extended-profile snapshot. Constant AWC does not mean a constant timing correction.

Determine AWC using only jointly measured support

The run-4399 reference is a fitted quadratic in inverse charge,

$$W_{\mathrm{MCP}}(Q) = a + b/Q + c/Q^2.$$

Let Ω be the intersection of the two outer profile-mean ranges, restricted to charge bins populated in both runs. Fit

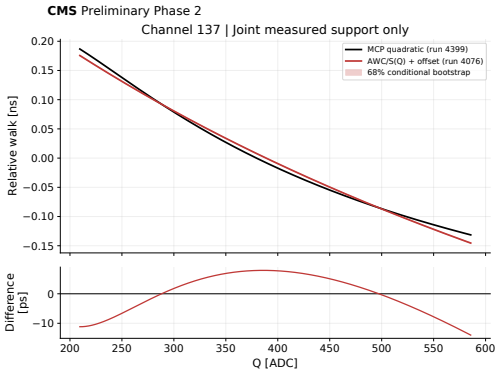
$$(A, B) = \arg\min_{A,B} \sum_{Q_k \in \Omega} \left[\frac{A}{c_0 + c_1 Q_k} + B - W_{\mathrm{MCP}}(Q_k) \right]^2 .$$

The saved implementation samples a uniform 3000-point charge grid before removing unsupported intervals. The offset B absorbs the arbitrary time zero.

This is uniform-charge shape matching, not an event-weighted likelihood. It uses neither extrapolated tails nor a single central anchor point.

Review N2-4: Compare weighting alternatives and propagate reference-fit uncertainty. The current objective is a documented choice, not a uniquely optimal estimator.

Channel 137: inverse-slope and MCP walk shapes



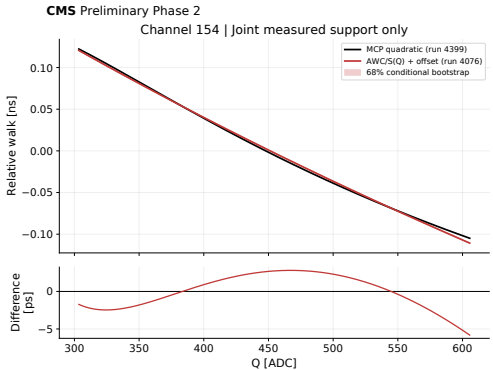
The black curve is the run-4399 MCP quadratic. The red curve is the run-4076 inverse-slope model after fitting AWC and a constant offset. Only jointly measured charge support is shown.

The lower panel is the difference between the two curves. Its RMS is 6.62 ps; this is not a single-event timing resolution.

The band varies the run-4076 slope fit while holding the MCP curve fixed.

AWC = 18.513 μA; conditional 16–84% interval [18.299, 18.791] μA. Both curves share one display offset.

Channel 154: a second shape-comparison example



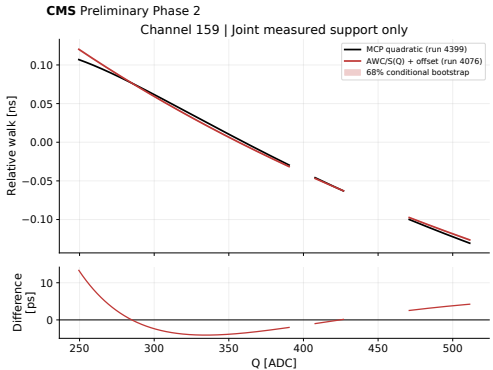
The same overlap-only objective determines AWC and the offset for channel 154. Both plotted curves are calibrated models, not matched events from different runs.

The lower panel directly displays the remaining shape mismatch, with no additional line subtracted.

Review N2-5: This replaces the ambiguous old residual illustration. It must be described as model closure, not a corrected-event resolution.

Conditional bootstrap band: 4076 slope replicas, refitted AWC and offset, fixed 4399 quadratic.

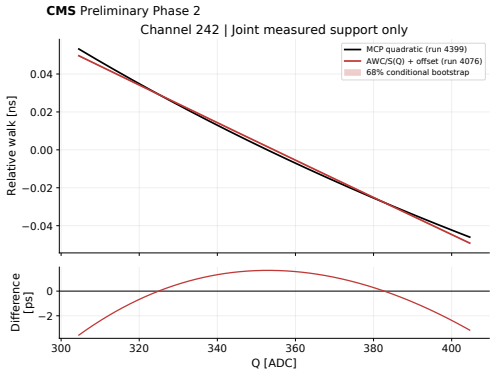
Channel 159: missing support is not extrapolation



Only intervals with populated profile bins in both runs contribute to the alignment. Breaks in the displayed curves mark excluded charge intervals. They do not represent discontinuities in either analytic model.
The lower panel uses exactly the same supported points. All displayed intervals participate in the common AWC-and-offset fit.

The stored comparison excludes 703 of the 3000 initial grid points for this channel. No line is drawn through those gaps.

Downstream example: channel 242

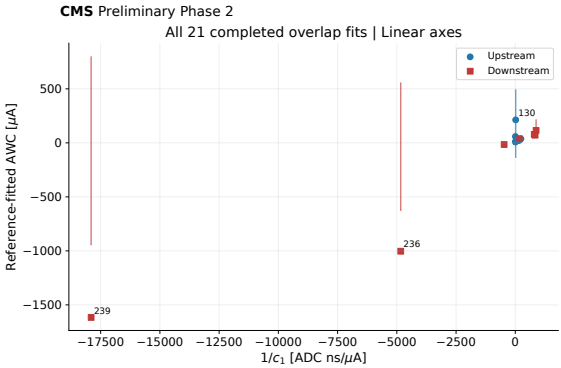


The method is also applied to a downstream channel. In run 4399, upstream bar 8 supplies the coincidence pair alongside the downstream target pair and MCP indicator.
In run 4076, the target downstream pair is required, without fixing an upstream pair. The curves use only their common measured support.

Review N2-1: Verify the transfer conditions and selection-induced changes for downstream channels as well as upstream channels.

The comparison is a per-channel shape calibration. The other module is not used as an internal timing reference to determine AWC here.

AWC versus inverse slope coefficient: all completed fits

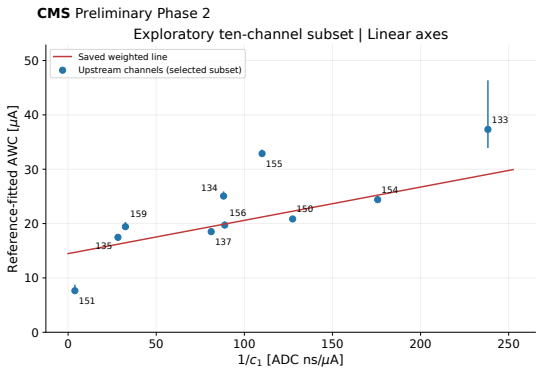


Each point is one channel, not one charge bin. Blue circles are upstream and red squares downstream. The x coordinate is $1/c_1$, the inverse coefficient of the linear $S(Q)$ fit; the y coordinate is reference-fitted AWC. Bars span the 16th–84th bootstrap percentiles. There are 21 completed fits; channel 238 has no measured overlap.

Review N2-6: Publish eligibility and failure criteria, including narrow support and poorly constrained channels.

Linear axes. Large intervals and negative coefficients are retained; this is not a final quality-selected performance sample.

A selected subset does not establish a universal law



The ten-channel upstream subset previously selected for discussion is shown on linear axes. Bars are conditional AWC bootstrap intervals; channel numbers identify the points. The saved weighted straight line has $\chi^2/\mathrm{ndf} = 607.7/8$ under its limited error model. It does not adequately describe the variation.

Review N2-6: The subset was selected interactively. It is exploratory, not an unbiased population-level validation.

The fit treats x as fixed and uses half the 16–84% AWC span as its symmetric uncertainty. Correlations and reference errors are omitted.

What the present uncertainties do and do not include

Displayed quantity	Meaning
Profile bars	SD of selected hits in each charge bin; not the uncertainty on the mean.
Fit weights	SEM of each profile mean; not the SD.
Slope band	16–84% range from 1000 bootstrap linear fits with fixed bin edges.
AWC bars / walk band	Propagate the slope replicas through a new AWC-plus-offset alignment to the same fixed reference curve.

Not yet included: run-4399 quadratic uncertainty, cross-run response transfer, selection/model choices and all shared-feature/target correlations.

Review N2-4: These are conditional statistical intervals, not total calibration uncertainties. A small curve mismatch is not a resolution estimate.

Result and next validation step

- A dual-threshold measurement provides a charge-dependent slope proxy; an inverse-slope function can be aligned to independent MCP walk curves.
- Restricting alignment to jointly measured charge support avoids determining AWC from extrapolation.
- Reference-calibrated AWC varies between channels. A straight line against $1/c_1$ is insufficient for the explored subset.
- The present calibration still uses the MCP to determine AWC. It does not establish reference-free AWC extraction by itself.
- An internal predictor must be tested on held-out bars/configurations with transfer and target uncertainties included before use without MCP.

The following backup explores such a predictor; it is not a final performance claim or a demonstration with cosmic rays.

Backup: an internal predictor for channel AWC

For $W = A/S$ and $S = c_0 + c_1 Q$,

$$-\frac{dW}{d\ln Q} = \frac{A\, c_1 Q}{S(Q)^2}.$$

If this local walk derivative is approximately shared between channels at a prescribed typical charge Q_*, a candidate model is

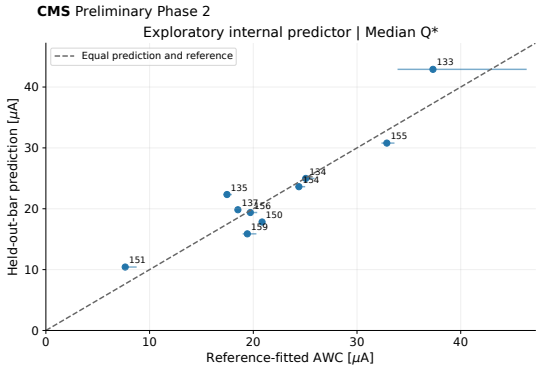
$$A_{\mathrm{pred}} = K\, \frac{S(Q_*)^2}{c_1 Q_*}.$$

Here Q_* is the median charge of the run-4076 quality- and peak-right-selected sample. It is computed once per channel; A_{pred} remains constant when correcting individual events.

The common K is learned from MCP-calibrated training channels. Predicting a new channel uses internal features, but the model development is reference-based.

Review N2-7: Shared local derivative is a hypothesis. Test operating-condition dependence and avoid selecting Q_* only after inspecting prediction scores.

Backup: predict a bar excluded from the model fit



Each point compares reference-fitted AWC with a prediction made after excluding that channel's entire bar from the fit of K. Ten upstream channels span six held-out bar groups.
The RMS prediction error is 3.02 μA; the RMS fractional error is 17.2%. Horizontal intervals are conditional reference-AWC intervals, not predictive intervals.

Review N2-7: Small selected sample; feature/target correlations and full prediction uncertainties remain unpropagated.

Exploratory median-Q_* candidate. The dashed line indicates equality, not a fitted result. No blanket 10% precision claim is made.

Review queue before publication

ID	Decision or missing evidence
N2-1	Freeze cross-run operating conditions and one compatible MCP-reference version.
N2-2	Document safe-t_2, threshold conversion and exact selected populations.
N2-3	Check linear-profile adequacy and selection/model sensitivity.
N2-4	Validate alignment objective and complete the uncertainty propagation.
N2-5	Keep model closure distinct from event timing-resolution validation.
N2-6	Freeze channel eligibility independently of observed trend; publish exclusions.
N2-7	Decide whether internal prediction remains backup or gains enough validation for a main result.

Sources: supplied `Note2.pdf`; extended profile snapshots and the dated measured-overlap study. All plotted tables and checksums accompany this draft.

External bibliography and publication metadata require a final audit. This review slide is removed from a public release.